

SQL Server Setup & Troubleshooting Guide

1 — Find SQL Server Instance Name

Check installed instances:

```
reg query "HKLM\SOFTWARE\Microsoft\Microsoft SQL Server" /v InstalledInstances
```

Update dbconfig.json based on result:

MSSQLSERVER → "server": "."

SQLEXPRESS → "server": ".\SQLEXPRESS"

SQLEXPRESS2022 → "server": ".\SQLEXPRESS2022"

List all instances on network:

```
sqlcmd -L
```

2 — Check SQL Server Service Status

```
sc query MSSQLSERVER
```

```
sc query MSSQL$SQLEXPRESS
```

```
sc query SQLBrowser
```

3 — Start / Stop Services

```
net start MSSQL$SQLEXPRESS
```

```
net stop MSSQL$SQLEXPRESS
```

```
net start SQLBrowser
```

4 — Test Connection

```
sqlcmd -S .\SQLEXPRESS -Q "SELECT name FROM sys.databases"
```

```
sqlcmd -S .\SQLEXPRESS -Q "SELECT db_id('RestaurantPOS')"
```

Returns a number → Database exists

Returns NULL → Database not created yet

```
sqlcmd -S .\SQLEXPRESS -Q "SELECT @@VERSION"
```

5 — Check Installation Logs

```
explorer "%ProgramFiles%\Microsoft SQL Server\160\Setup Bootstrap\Log"
```

```
explorer "%TEMP%"
```

6 — Enable Protocols

```
mmc SQLServerManager16.msc
```

7 — dbconfig.json Location

```
explorer "%LOCALAPPDATA%\RestaurantPOS"
```

```
type "%LOCALAPPDATA%\RestaurantPOS\dbconfig.json"
```